

ALEX CUGNOD SEGOVIA

Data Scientist - Statistician - Machine Learning

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PROFILE

Statistician and Data Scientist with hands-on experience building data-driven solutions for industrial manufacturing at Renault Group. Strong background in statistical modeling, machine learning, predictive analytics and data visualization, with end-to-end experience turning fragmented business data into practical analytical applications. Experienced in Python, FastAPI, statistical modeling and time series, with a strong focus on solving open-ended problems independently.

PROFESSIONAL EXPERIENCE

Data Scientist & Software Engineer Renault Group - Innovation Department | Palencia, Spain Jan 2026 - Present

Absenteeism Analytics & Prediction Platform:

- Built an end-to-end absenteeism analytics platform integrating React, FastAPI, Python, Looker, local datasets and web/JSON sources, working with 3M+ person-day records across 99 variables and growing.
- Developed the application architecture from data processing and API services to an interactive React frontend, replacing fragmented Excel/legacy-tool workflows with a centralized analytical platform.
- Built 10+ functional modules covering factory, department, workshop, UET and employee-level analysis, with cascading filters and interactive data exploration.
- Implemented authentication, sessions, role-based access control, employee management, configuration management and automated reporting through a FastAPI backend and React frontend.
- Developed interactive analytical dashboards with Plotly, including absenteeism evolution, organizational breakdowns, employee tracking, recurrence analysis and operational KPIs.
- AI assistant: integrated a natural-language assistant capable of querying absenteeism data within the user's current organizational scope and returning tables, charts and employee-level results.
- Forecasting: developed experimental absenteeism forecasting models using Elastic Net, Random Forest and SARIMAX, with calendar features, lagged variables and temporal validation.
- Designed a walk-forward validation framework from scratch to evaluate forecasting performance while avoiding temporal data leakage.
- Worked with managers, HR and operational stakeholders to identify data-quality issues and define consistent absenteeism metrics and business rules.

Production Analytics & Optimization:

- Maintained and extended Docuzilla, a Python/DuckDB tool used daily by Renault engineers to identify similar vehicle configurations and investigate production issues.
- Optimized core processing logic, reducing execution time to less than 10% of the original runtime, while developing new features and algorithmic improvements.
- Developed an experimental next-day factory gas-consumption forecasting model using historical consumption, production and weather data.
- Designed a PostgreSQL database for quality-management dossiers, including schema, relationships, constraints, indexes and views.

Data Scientist / Analyst ITACyL - Agricultural Technological Institute of Castilla y Leon | Valladolid, Spain Apr 2025 - Jul 2025

- Developed the institute's first predictive modeling approach for common vole population dynamics using historical regional data through 2024.
- Built statistical and machine learning models in R and Python with spatial and temporal validation; reduced 145 candidate variables to 12 key predictors.
- Achieved $R^2 = 0.75$ in geographical validation and >95% accuracy in forecasting population trend direction.
- Identified demographic cycles ranging from approximately 2 to 7 years.

SELECTED PERSONAL PROJECTS

SignalDesk Python, FastAPI, Finance, Machine Learning Demo: <https://signaldesk-q9n7.onrender.com/> Research tool for prioritizing insider-trading filings, combining

SEC Form 4 data, market context and anomaly detection into quantitative BUY/SELL/HOLD/IGNORE signals with confidence scores - plus a public transparency track record benchmarked against the S&P 500.

Earnings Call NLP Pipeline Python, FastAPI, FinBERT, NLP Demo: <https://earnings-call-nlp.onrender.com/> | GitHub: <https://github.com/aleex-cs/earnings-call-nlp>

Dashboard that scores SEC 8-K Exhibit 99.1 earnings releases with FinBERT, tracks uncertainty language over time, compares issuers, and overlays next-session stock returns versus SPY.

Music Stats Python, Streamlit Demo: <https://musicstats.streamlit.app/> | GitHub: https://github.com/aleex-cs/music_stats

Listening-habits dashboard built on Last.fm scrobble history, with sunburst charts, streamgraphs, animated bar-chart races and automatic listening-session detection.

DataPipe (Work in Progress) FastAPI, React, D3.js, Static Analysis Code analysis and visualization tool that turns a zipped Python project into an interactive graph of its data flow, showing every read, transformation and processing step from source to output.

Literary Character Stylometry (Work in Progress) Python, NLP, LLM, spaCy NLP/LLM pipeline that compares literary characters across books to find psychological and narrative similarities, using a local LLM for profile generation and multilingual embeddings for similarity scoring.

WatchList (Work in Progress) Python, JavaScript, Supabase, TMDB API Full social web app for tracking films and shows, with authentication, episode-by-episode tracking, detailed viewing statistics, awards tracking and a bulk-import system from spreadsheets.

Movie Stats (Work in Progress) Python, Streamlit Three-year personal project analyzing ~900 films using a curated/scraped dataset, with interactive statistical analyses and visualizations.

Financial Portfolio Optimization R, Finance, Markowitz Model, Statistics GitHub: <https://github.com/aleex-cs/financial-portfolio-optimization>

Analysis and optimization of a seven-stock portfolio (2020-2025): efficient frontier, robust covariance estimation, and comparison of minimum-variance, maximum-return, maximum-Sharpe-ratio and equal-weighted portfolios.

EDUCATION

BSc in Statistics University of Valladolid (UVA) | Valladolid, Spain Sep 2021 - Jul 2025 Thesis grade: 9.8 / 10. Thesis: "Predictive Models for Common Vole Population Fluctuations and Agricultural Crop Risk" - early-warning modeling using time-series analysis.

TECHNICAL SKILLS

Programming: Python, R, SQL, JavaScript

Data Science & Statistics: Regression, GLM, LASSO, Elastic Net, Random Forest, Gradient Boosting, SARIMAX, ETS, STL, PCA, Clustering, Bayesian Methods, Feature Engineering, Time Series,

NLP, PyTorch

Data & Visualization: Pandas, NumPy, Scikit-learn, Plotly, Power BI, Looker, Streamlit

Web & APIs: React, FastAPI, REST APIs, Vite, React Router, HTML/CSS

Databases & Tools: PostgreSQL, DuckDB, Supabase, Git, AWS, Azure

Other: FinBERT, Hugging Face, AI Agents, RAG, Data Pipelines, Web Scraping

CERTIFICATIONS & TRAINING

Renault Group: Looker SSBD Training

LinkedIn Learning:

- Agentic AI: Building Data-First AI Agents

<https://www.linkedin.com/learning/certificates/2b9aa83b226385c3b98be14f3d1f709c41ba341da7576b73d2c45252afe4388f>

- Build with AI: Autonomous Agents with LangChain and Hugging Face

<https://www.linkedin.com/learning/certificates/08f4a6c5994d211d9ed641870e6ef263fa5ce743c299e3a916ccd00ca31f6ab2>

- Building a Personalized Chatbot with OpenAI and LangChain

<https://www.linkedin.com/learning/certificates/d58803a73453eb3a21fb0d6e04cb1178cfa47a0e417007651b1c65092bd8222a>

- Program Evaluation for Data Science

<https://www.linkedin.com/learning/certificates/308e4f684573e60adc641f576f9d6640e25de20a0b2cf4c1c25c62671a5c8b8b>

- Advanced NoSQL for Data Science

<https://www.linkedin.com/learning/certificates/9b2c5bcb25de3952842363b87d5f2b2396f9123955e2969742671da3a38d4220>

- AWS Certified Machine Learning - Specialty (MLS-C01) Cert Prep

<https://www.linkedin.com/learning/certificates/c6980b87f040dd6cce36d2004715f9fae47e43a742152da339a9abcda7a7b52f>

LANGUAGES

Spanish: Native English: Full Professional Proficiency Italian: Upper-Intermediate / Advanced